

279kWh **Smart** Battery Cabinet



279kWh

ENERGY
CAPACITY

125kW

GUARANTEED
MIN POWER

821V

NOMINAL
VOLTAGE

6K+

CYCLE LIFE
@ 80% DOD

-40/+55°C

OPERATING
TEMP

IP54

ENCLOSURE
RATING

20yr

DESIGN
LIFE

Smart

ENERGIOS™
ENABLED

Series B 279kWh Smart Battery



- Sodium-Ion NFPP
- Air-Cooled Cabinet
- EnergiOS™ Smart EMS
- IP54 Outdoor-Rated



Built for critical infrastructure resiliency and demanding commercial & industrial use cases, the Series B 279kWh Smart Battery Cabinet prioritizes compliance, field-validated performance, and bankable warranties — delivering a **20-year design life** and **6,000+ cycles at 80% DoD** with zero maintenance and no HVAC required.

INCLUDES AN OUTDOOR, IP54-RATED ENCLOSURE CONTAINING:

- Sodium-Ion NFPP battery modules
- Air-cooled thermal management
- Auxiliary safety systems
- Battery Management System (BMS)
- EnergiOS™ Smart EMS platform
- Fire suppression & gas detection

KEY BENEFITS



Safety-First Design

Sodium-Ion NFPP chemistry eliminates thermal runaway risk — enabling safer siting, simpler AHJ approvals, and no fire suppression complexity.

- ✓ Zero thermal runaway incidents
- ✓ No toxic gas emissions
- ✓ Air-shippable — faster deployment
- ✓ Cleaner AHJ approvals



Extreme Performance

Operates from -40°C to +55°C without derating or HVAC infrastructure — ideal for outdoor installations, cold storage, and industrial environments.

- ✓ -40°C to +55°C operation
- ✓ No HVAC required
- ✓ Minimal capacity derating
- ✓ Fast charging capability



Intelligent & Bankable

Built-in EnergiOS™ Smart EMS transforms the cabinet into an active energy asset — optimizing dispatch, enabling VPP participation, and maximizing ROI.

- ✓ 6,000+ cycles @ 80% DoD
- ✓ 20-year design life
- ✓ ITC eligible with solar PV
- ✓ Demand response & VPP ready

WHERE THE SERIES B PERFORMS



MANUFACTURING / COLD STORAGE / DISTRIBUTION

Peak Demand Management

Reduce demand charges 30–50% at manufacturing facilities, cold storage, and distribution centers while maintaining full capacity in extreme temperatures.



DATA CENTERS / HEALTHCARE / TELECOM

Critical Infrastructure

Guaranteed uptime for data centers, hospitals, and telecom — with air-cooled simplicity, intelligent monitoring, and no fire suppression complexity.



WAREHOUSE / PORT / AIRPORT

Material Handling & Fleet

Fast charging and extreme temp tolerance for 24/7 operations — warehouse automation, port equipment, and airport ground support.



SOLAR / WIND / CAMPUS / COMMUNITY

Renewable Integration

Long-duration storage with 20-year design life for bankable solar + storage projects, campus microgrids, wind smoothing, and community solar.

THREE TECHNOLOGIES. ONE INTELLIGENT PLATFORM.

Not Just a Cabinet — An Intelligent Energy Platform

The Series B integrates with Coulomb Technology's Smart EMS platform — a suite of three proprietary software technologies that transform your battery cabinet into an active, revenue-generating energy asset. Monitor, optimize, and control your energy in real time from any device.



The intelligent operating system for your energy storage. Provides real-time monitoring, automated charge/discharge scheduling, and seamless solar integration.

- Real-time SOC, power flow & health monitoring
- Automated TOU & solar charge scheduling
- Remote access via web & mobile app
- Alerts, diagnostics & OTA firmware updates



Machine learning algorithms that learn energy usage patterns, weather forecasts, and utility rates to autonomously optimize dispatch and maximize savings.

- Predictive load forecasting & solar yield modeling
- Autonomous dispatch optimization
- Dynamic rate arbitrage (TOU, demand charges)
- Continuous learning — improves over time



Advanced grid interaction capabilities enabling virtual power plant participation, demand response programs, and utility incentive monetization.

- Virtual power plant (VPP) participation
- Demand response & utility incentive programs
- Grid frequency regulation support
- Export control & islanding management

INTELLIGENT BMS PROTECTIONS



Overcharge Protection



Overcurrent Protection



Overdischarge Protection



Overheat Protection



Temperature Management



Short Circuit Protection



Balanced Power

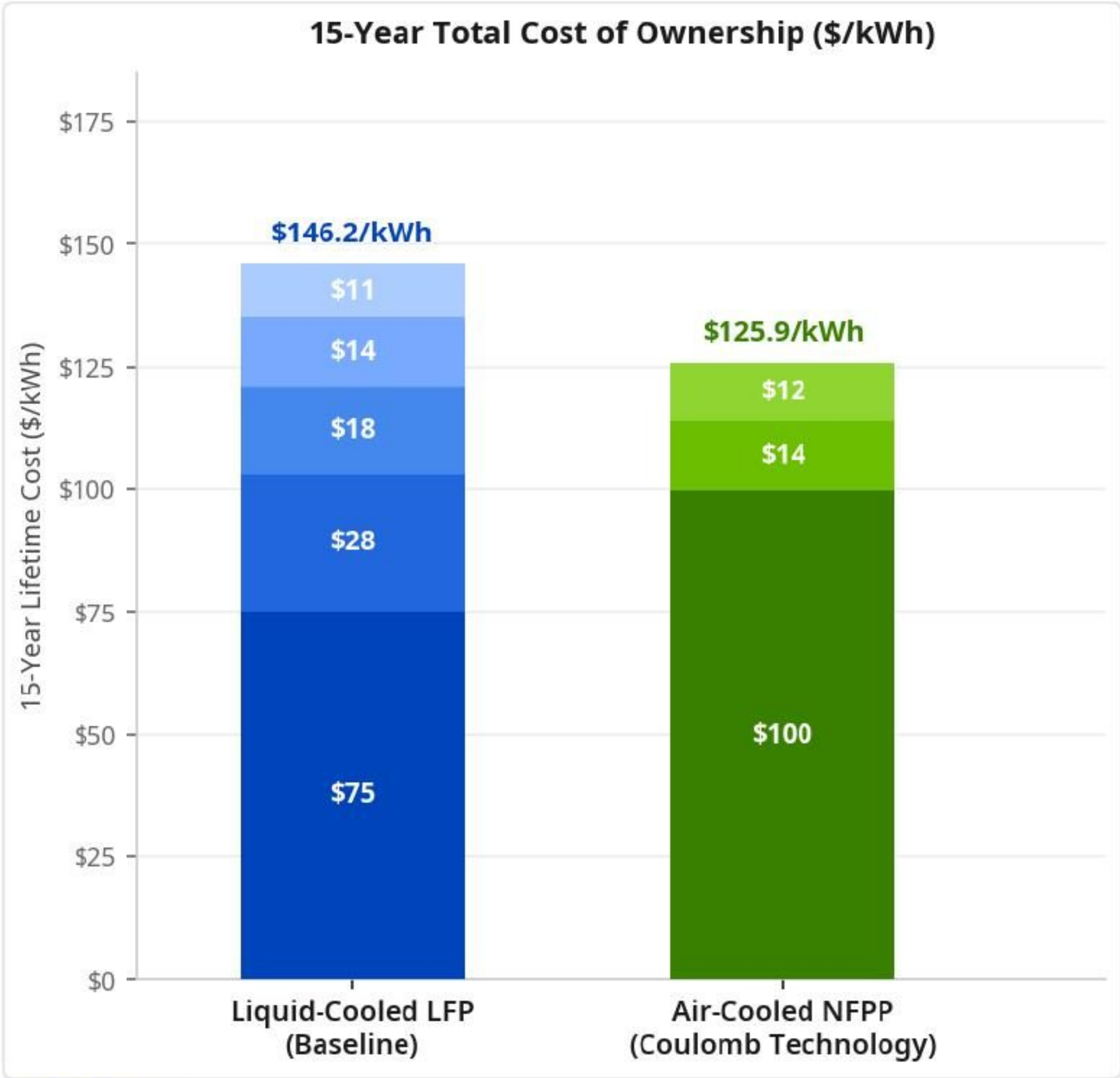


Reverse Connection

SYSTEM ADVANTAGE OVER LFP-BASED SYSTEMS

14% Lower Total Cost of Ownership Over the Project Lifetime

Our sodium-ion technology delivers measurable savings through eliminated HVAC requirements, reduced degradation, and lower maintenance, insurance, and permitting costs — validated by real-world deployments, not laboratory projections.



MODEL ASSUMPTIONS	
NFPP installed price delta	\$16/kWh
Cycles per year	365
HVAC load delta	90%
Insurance cost delta	20%
Augmentation delta	20%
O&M cost delta	30%
Cell derating / cold temps	75%
Discount rate / NPV calc	8%

COST COMPONENTS	
LFP — CapEx	NFPP — CapEx
LFP — HVAC Load	NFPP — O&M
LFP — O&M	NFPP — Insurance
LFP — Insurance	LFP — Augmentation

14% TCoO Savings — Air-cooled NFPP (\$125.9/kWh) vs. Liquid-cooled LFP (\$146.2/kWh) over a 15-year project lifetime.

\$20.3/kWh — Lifetime savings per kWh over project lifetime

20yr — Projected battery life, sodium-ion NFPP chemistry

RESEARCH & DEVELOPMENT PARTNERS



SERIES B — 279KWH
CABINET



ENERGY
CAPACITY

279kWh

229 kWh usable at 85% DOD

DURATION RANGE

2.2–10hr

Peak shaving to long backup

MAX RATED POWER

140kW

125 kW guaranteed constant

OPERATING TEMPERATURE

–40/+55°C

No HVAC — any climate

DC VOLTAGE RANGE

634–994v

Flexible inverter compatibility

ENCLOSURE RATING

IP54

Outdoor-rated direct install

PARAMETER	VALUE / DESCRIPTION	UNIT
ENERGY CAPACITY	279	kWh
USABLE ENERGY (85% DOD)	229	kWh
MAX RATED POWER	140	kW
GUARANTEED CONSTANT POWER	125	kW
DC VOLTAGE RANGE	634 – 994	VDC
DURATION	2.2 – 10	Hours
CYCLE LIFE	6,000+ at 80% DoD, 25°C	—
CELL TYPE	Sodium-Ion NFPP (Non-Flammable Polymer Pouch)	—
OPERATING TEMP (DISCHARGE)	–40°C to +55°C	—
CHARGING TEMP RANGE	–10°C to +45°C	—
DESIGN LIFE	20 years	—
ENCLOSURE (IP RATING)	Outdoor Cabinet — IP54	—
THERMAL MANAGEMENT	Air-Cooled (No HVAC Required)	—
BMS COMMUNICATION	Integrated BMS, Modbus TCP / CAN Bus	—

CERTIFICATIONS

- UL 1973
- UL 9540
- UN 38.3
- IEC 62619

Cell & Module Level

UL 9540A TESTING

Cell: Tested & Passed
Module/System: In Progress

WARRANTY

Cell: 3 Years
System: 5 Years
Extended warranty available

FACTORY ACCEPTANCE TEST (FAT) SCOPE

- ✓ Visual, Mechanical & Torque Inspection

✓ 3 × 0.125C Capacity/Energy Verification (85% DOD)
- ✓ Protections & Isolation Tests

✓ Baseline OCV Measurement
- ✓ Functional & Conditioning Test

✓ 10 × 0.5C Stability Cycles with Power Checks

WHY SODIUM-ION NFPP OUTPERFORMS LEGACY LEAD-ACID TECHNOLOGY

A Clear Advantage Across Every Performance Dimension

Coulomb Technology's Sodium-Ion NFPP chemistry delivers superior cycle life, wider operating temperature range, zero maintenance, and a fundamentally safer chemistry — making it the ideal replacement for lead-acid in critical commercial and industrial applications.

FEATURE	SODIUM-ION NFPP	LEAD-ACID
Cycle Life	3,000+ cycles @ 80% DoD	300–500 cycles @ 50% DoD
Weight (100Ah)	13 kg (28.7 lb)	~22 kg (48.5 lb)
Usable Capacity	80–100% DoD	50% DoD (to preserve life)
Discharge Temp Range	–40°C to +60°C	–20°C to +50°C
Charge Temp Range	–10°C to +45°C	0°C to +40°C
Maintenance	Zero maintenance	Regular watering & equalization
Self-Discharge	<3% per month	5–15% per month
Charging Efficiency	~95%	~75%
Thermal Runaway Risk	Zero risk	Low risk
Toxic Gas Emissions	None	Hydrogen gas during charging
Memory Effect	None	Sulfation if undercharged
Lifespan	10+ years	3–5 years
Environmental Impact	Non-toxic, recyclable	Lead & acid hazardous

30%

Safer chemistry
vs. LFP lithium-ion

50%

More extreme temp
tolerant vs. LFP

20%

Less degradation
vs. LFP lithium-ion

Zero

Thermal runaway
risk